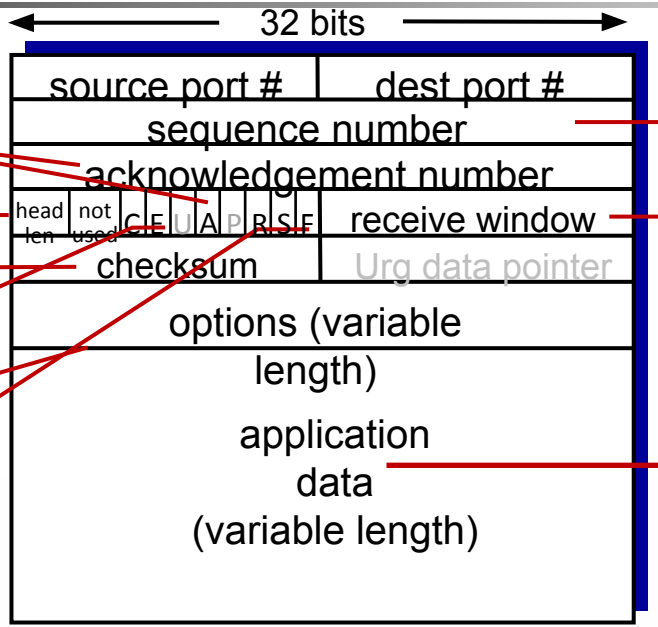
Chapter 3: Roadmap

- Transport-layer services
- Multiplexing and demultiplexing
- Connectionless transport: UDP
- Principles of reliable data transfer
- Connection-oriented transport: TCP
  - segment structure
  - reliable data transfer
  - flow control
  - connection management
- Principles of congestion control
- TCP congestion control

TCP: overview RFCs: 793,1122, 2018, 5681, 7323

- point-to-point:
  - one sender, one receiver
- reliable, in-order *byte stream:*
  - no “message boundaries"
- full duplex data:
  - bi-directional data flow in same connection
  - MSS: maximum segment size
- cumulative ACKs
- pipelining:
  - TCP congestion and flow control set window size
- connection-oriented:
  - handshaking (exchange of control messages) initializes sender, receiver state before data exchange
- flow controlled:
  - sender will not overwhelm receiver

TCP Segment Structure



ACK: seq # of next expected byte; A bit: this is an ACK

length (of TCP header)

Internet checksum

C, E: congestion notification

TCP options

RST, SYN, FIN: connection management

segment seq #: counting bytes of data into bytestream (not segments!)

flow control: # bytes receiver willing to accept

data sent by application into TCP socket

TCP sequence numbers, ACKs

Sequence numbers:

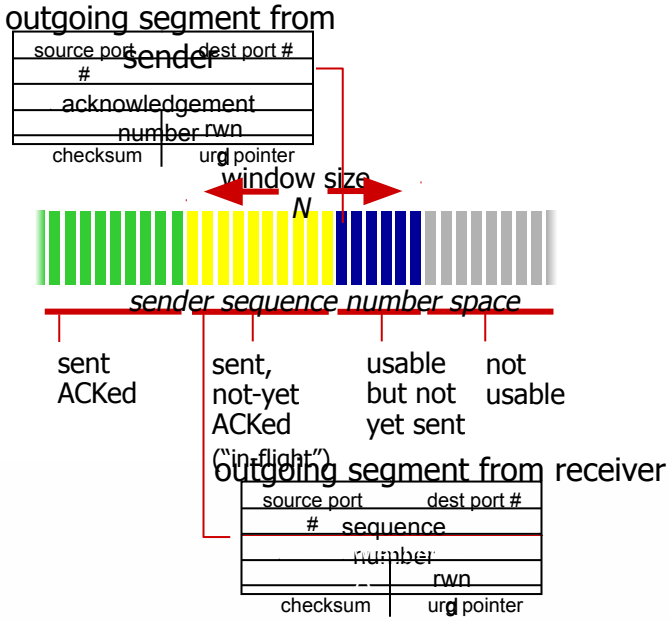
- byte stream "number" of first byte in segment's data

Acknowledgements:

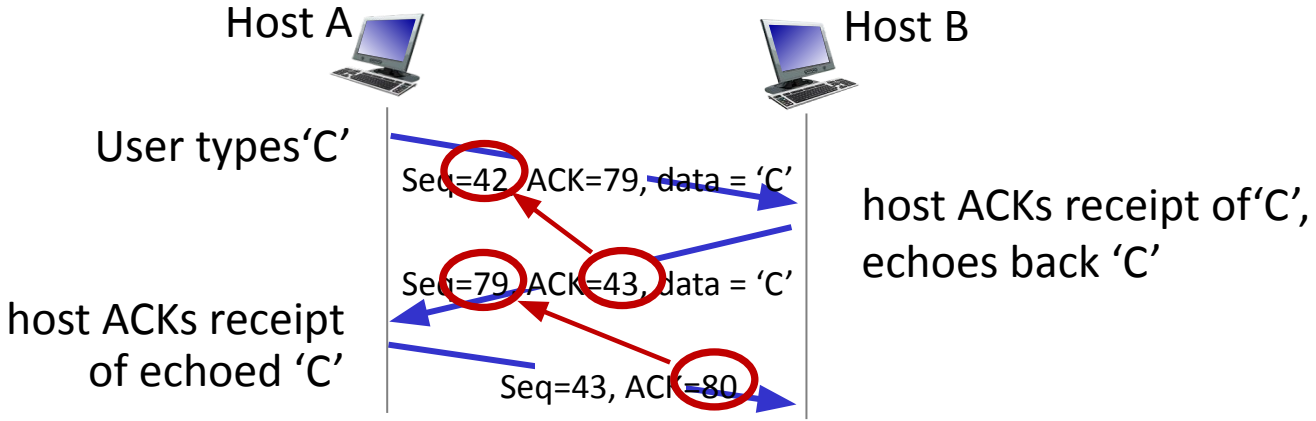
- seq # of next byte expected from other side
- cumulative ACK

Q: how receiver handles out-of-order segments

- *A:* TCP spec doesn't say, - up to implementor



TCP sequence numbers, ACKs



simple telnet scenario

TCP round trip time, timeout

Q: how to set TCP timeout value?

- ✔ longer than RTT, but RTT varies!
- ✔ *too short:* premature timeout, unnecessary retransmissions
- ✔ *too long:* slow reaction to segment loss

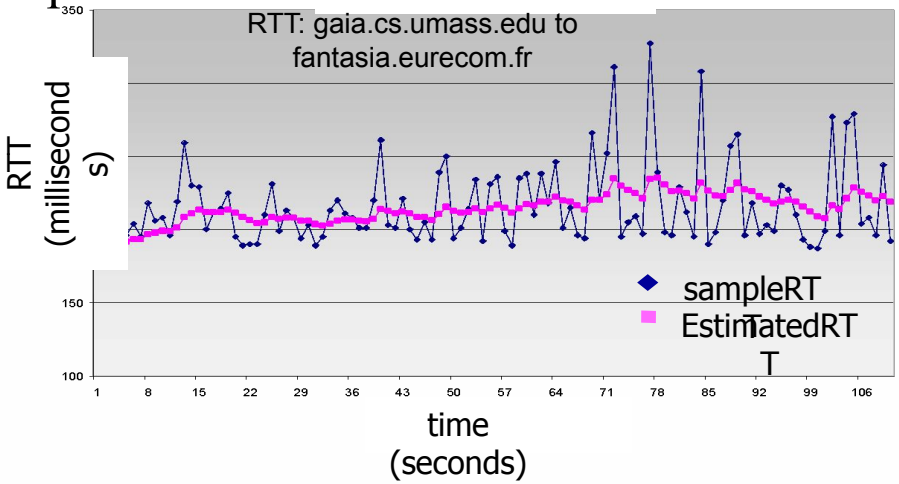
Q: how to estimate RTT?

- SampleRTT: measured time from segment transmission until ACK receipt
  - ✔ ignore retransmissions
- SampleRTT will vary, want estimated RTT "smoother"
  - ✔ average several *recent* measurements, not just current SampleRTT

TCP round trip time, timeout

EstimatedRTT = (1- α)*EstimatedRTT + α*SampleRTT

- exponential weighted moving average (EWMA)
- influence of past sample decrea exponentially fast
- typical value: α = 0.125



TCP round trip time, timeout

- timeout interval: **EstimatedRTT** plus "safety margin"
  - ✔ large variation in **EstimatedRTT:** want a larger safety margin

TimeoutInterval = EstimatedRTT + 4*DevRTT

estimated RTT

"safety margin"

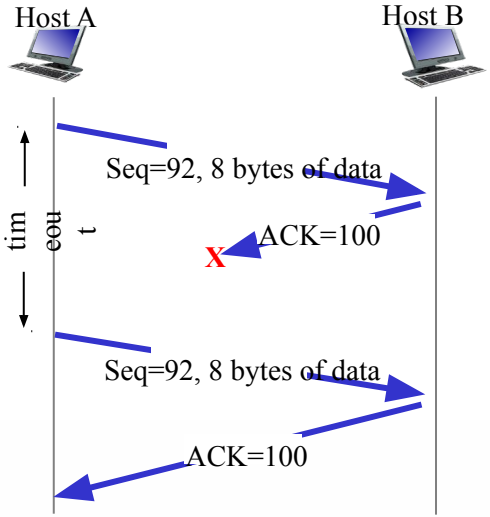
- **DevRTT**: EWMA of **SampleRTT** deviation from **EstimatedRTT**:

DevRTT = (1-β)*DevRTT + β*|SampleRTT-EstimatedRTT|

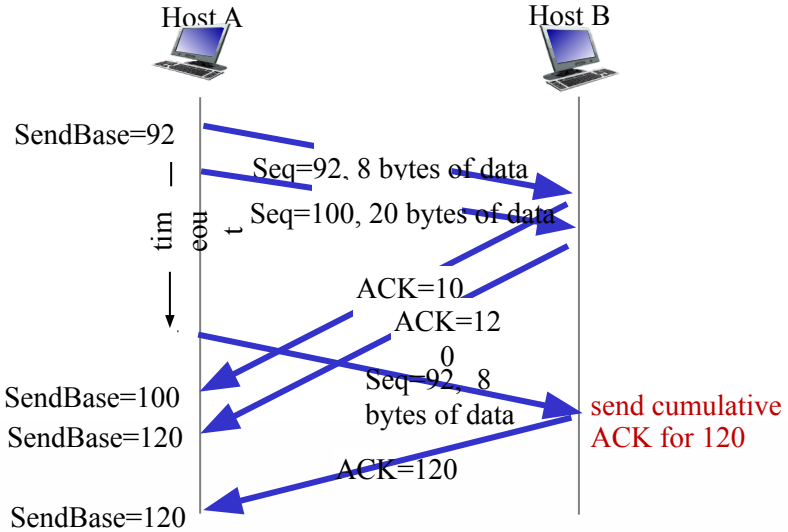
(typically, β = 0.25)

* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/

TCP: retransmission scenarios

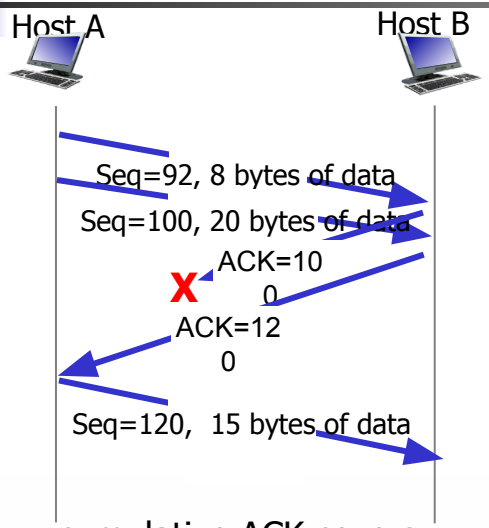


lost ACK scenario



premature timeout

TCP: retransmission scenarios



cumulative ACK covers for earlier lost ACK

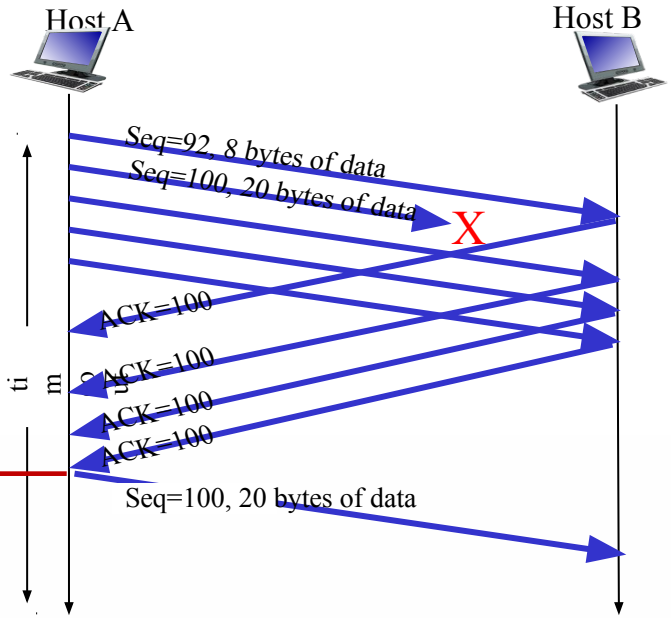
TCP Fast Retransmit

TCP fast retransmit

3 additional ACKs for same data ("triple duplicate ACKs"), resend unACKed segment with smallest seq #

- likely that unACKed segment lost, so don't wait for timeout

Receipt of three duplicate ACKs indicates 3 segments received after a missing segment – lost segment is likely. So retransmit!



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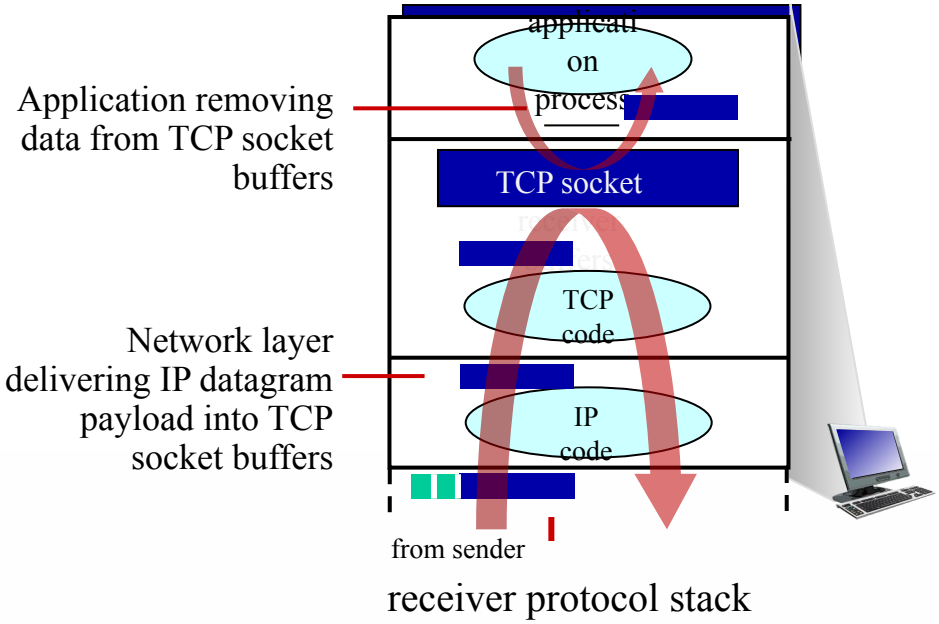
TCP Flow Control

Q: What happens if network layer delivers data faster than application layer removes data from socket buffers?



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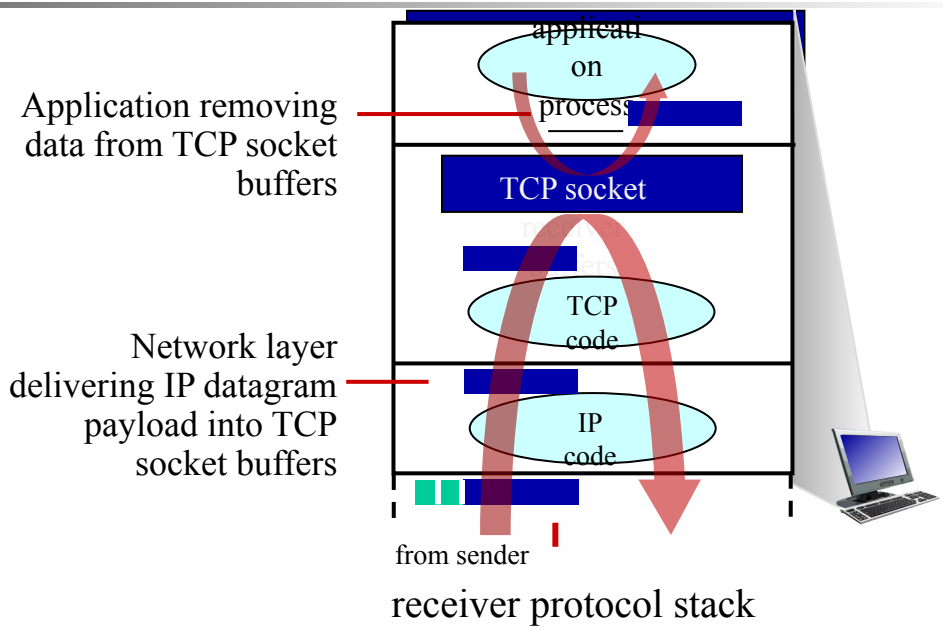

TCP Flow Control

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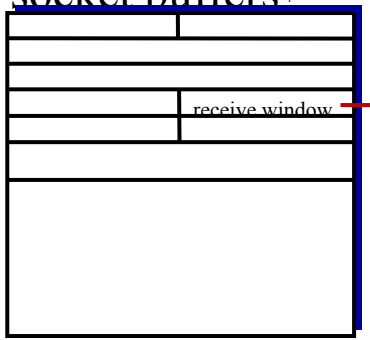
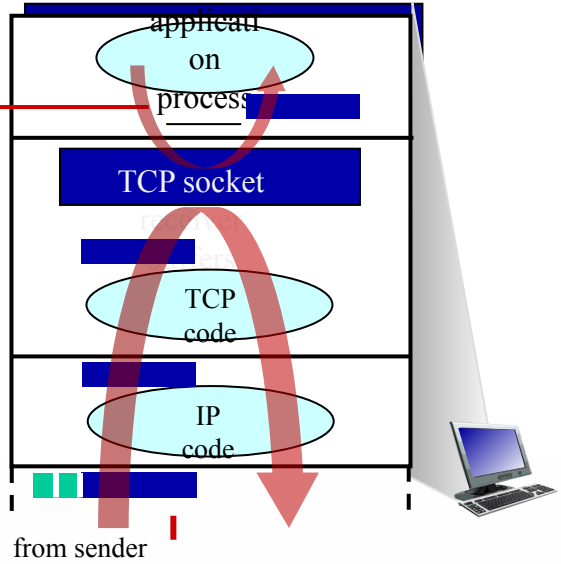


flow control: # bytes receiver willing to accept

Application removing data from TCP socket buffers



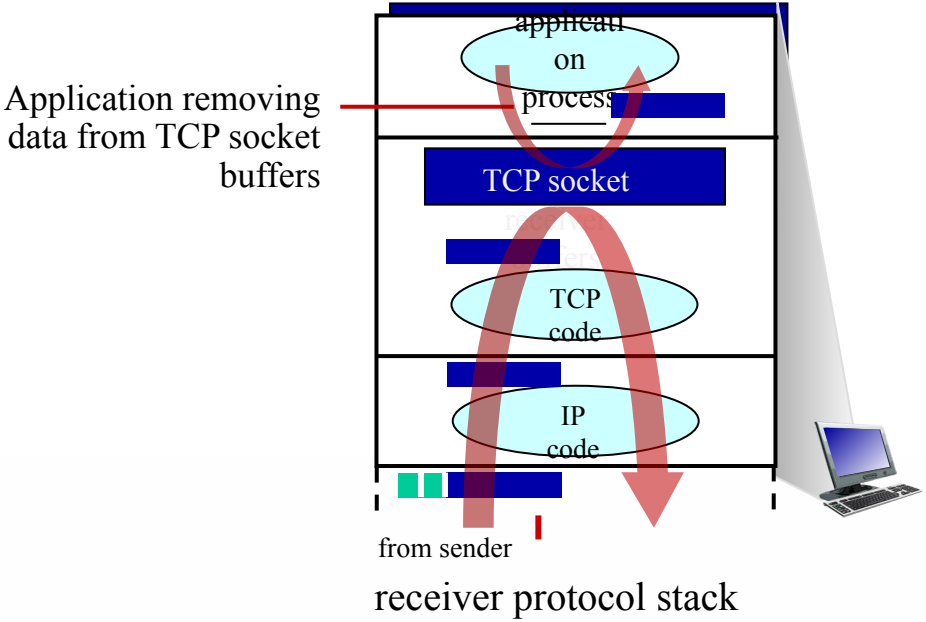
receiver protocol stack

TCP flow control

Q: What happens if network layer delivers data faster than application layer removes data from

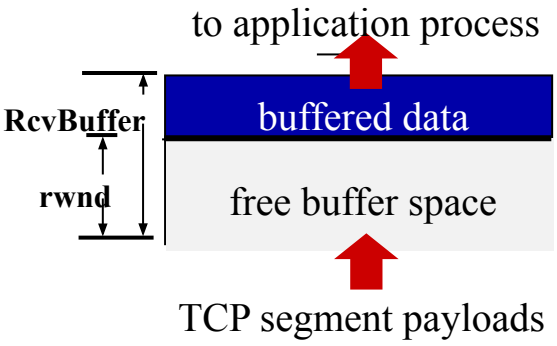
flow control

receiver controls sender, so sender won't overflow receiver's buffer by transmitting too much, too fast



TCP Flow Control

- TCP receiver "advertises" free buffer space in **rwnd** field in TCP header
  - **RcvBuffer** size set via socket options (typical default is 4096 bytes)
  - many operating systems autoadjust **RcvBuffer**
- sender limits amount of unACKed ("in-flight") data to received **rwnd**
- guarantees receive buffer will not overflow

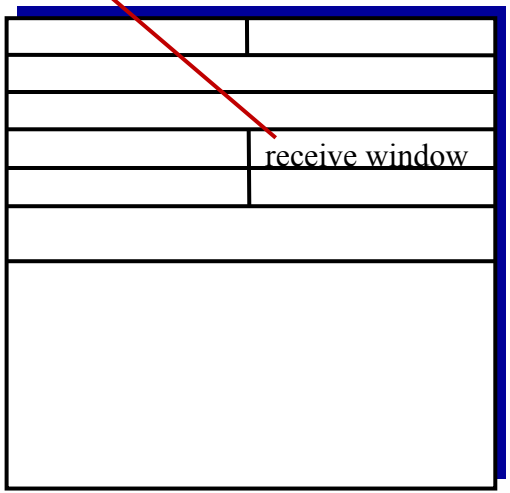


TCP receiver-side buffering

TCP Flow Control

- TCP receiver "advertises" free buffer space in **rwnd** field in TCP header
  - ✔ **RcvBuffer** size set via socket options (typical default is 4096 bytes)
  - ✔ many operating systems autoadjust **RcvBuffer**
- sender limits amount of unACKed ("in-flight") data to received **rwnd**
- guarantees receive buffer will not overflow

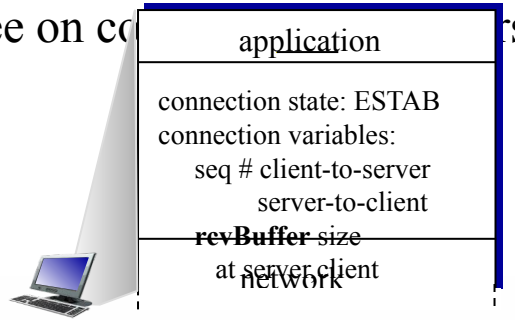
flow control: # bytes receiver willing to accept

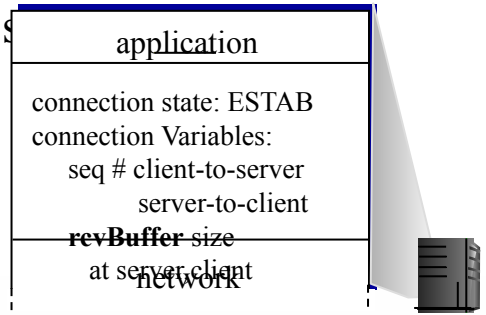


TCP segment format

TCP Connection Management

- before exchanging data, sender/receiver "handshake":
  - ✔ agree to establish connection (each knowing the other willing to establish connection)
  - ✔ agree on co rs (e.g., starting s



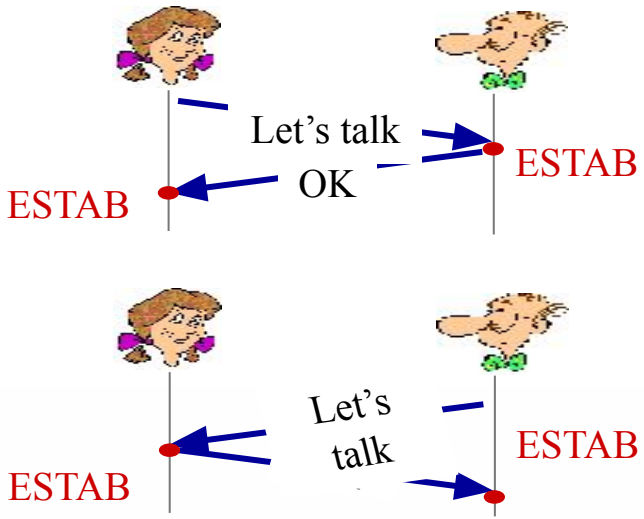


Socket clientSocket =
newSocket("hostname","port number");

Socket connectionSocket =
welcomeSocket.accept();

Agreeing to Establish a Connection

2-way handshake:



Q: TCP is bidirectional, will 2-way handshake work?

A: 2-way handshake is required for both the entities (i.e., client and server)

Both entities need to establish Initial Sequence Number (ISN)

TCP 3-way handshake

Client state

```
clientSocket = socket(AF_INET, SOCK_STREAM)
```

LISTEN

```
clientSocket.connect((serverName,serverPort))
```

SYNSENT

ESTAB

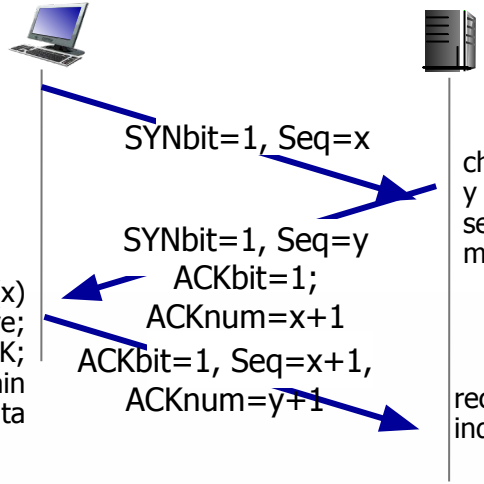
Server state

```
serverSocket = socket(AF_INET,SOCK_STREAM)
serverSocket.bind(('',serverPort))
serverSocket.listen(1)
connectionSocket, addr = serverSocket.accept()
```

LISTEN

SYN RCVD

ESTAB



Closing a TCP Connection

- client, server each close their side of connection
  - send TCP segment with FIN bit = 1
- respond to received FIN with ACK
  - on receiving FIN, ACK can be combined with own FIN
- simultaneous FIN exchanges can be handled

Thank You